Negin Mohajeri

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Amherst, MA

Research Interests: Mixed-Signal Integrated Circuit Design, Hardware Design for Neuromorphic Systems, Low Power Design, Hardware Security.
Objective: Seeking a Fall 2026/Spring 2027 Internship

Education

Ph.D. in Electrical and Computer Engineering Expected 2029
University of Massachusetts, Amherst, United States

Ongoing Research: Design and Integration of Transimpedance Amplifiers into Memristive Crossbar Arrays
This design is part of the project *Analog-Edge Processor Security* funded by ARL
Advisor: Dr. Wayne Burleson

M.Sc. in Electrical Engineering - Digital Electronic Systems 2020
Islamic Azad University - Science and Research Branch, Tehran, Iran

Thesis: Design and Simulation of High Performance, Low Power, and Robust Memristive Neural Network in the presence of Process Variations in Sub 90nm Technology
Advisor: Dr. Behzad Ebrahimi

B.Sc. in Electrical Engineering - Electronics 2014
Islamic Azad University - Central Tehran Branch, Tehran, Iran

Capstone Project: RFID Authentication and Access Control System for EE Campus
Advisor: Dr. Behzad Jalaei

Key Coursework

- VLSI design
- Analog Integrated Circuits
- Semiconductor Devices
- Security Engineering

Relevent Experience

Graduate Research Assistant *Feb 2025 – Present*
University of Massachusetts Amherst

- Designing and simulating low-power amplifiers to analyze and optimize circuit performance and scalability in memristive crossbar arrays and neuromorphic systems
- Developed and simulated a 64-bit XOR-arbiter PUF in HSPICE to evaluate robustness under process variations through challenge response pairs.

Embedded Software Engineer *May 2023 - Dec 2024*
Fanavaran DoorAndish Eshraq, Tehran, Iran

- Designed a far field and near field audio capture system using MEMS microphone arrays
- Implemented GPIO and serial communication protocols I^2C and SPI on the OMAPL138 SoC to interface keypads and voice processor FM1288 in an embedded communication system
- Designed a windows-based circuit testing application to improve hardware validation speed and efficiency

- Configured OpenWrt to connect radios without static firmware limitation

Hardware Test Engineer *April 2022 - May 2023*

Fanavaran DoorAndish Eshraq, Tehran, Iran

- Created test flow diagrams and comprehensive documentation for the production team to identify high risk hardware defects
- Identified critical design errors to improve test efficiency

Hardware Design Engineer *Nov 2020 - Sep 2021*

Arman Faraz Peyman, Tehran, Iran

- Designed multilayer (2- and 4-layer) PCB and schematic designs for elevator control circuitry
- Improved the Uninterruptible Power Supply(UPS) to ensure reliable operation of the elevator driver stage during power outages

Co-Founder, Hardware Design Engineer *June 2020 - Nov 2020*

KitLime, Tehran, Iran

- Developed hardware design strategies aligned with cost, scalability, and market requirement to support commercialization goals

Hardware Design Engineer *June 2018 - June 2020*

Medaria , Tehran, Iran

- Built a patient-protection system to enhance device safety during power outages and output frequency detection
- Developed a patient-contact detection circuit that improves device power efficiency
- Implemented an STM32F030F4-based board, integrating a watchdog supervisory circuit to automatically reset the device during power outages

Mentoring and Teaching Experience

Leadership Toolkits *May 2026 - June 2026*

University of Massachusetts Amherst

Project Mentor, Flipping Pancake Robot *July 2026*

SENGI , University of Massachusetts Amherst

- Mentored high school students through the Summer Engineering Institute for High School Students on a robotics project

Research Mentor *June 2026 - July 2026*

Student Research Institute (SRI), Harvard Undergraduate OpenBio Labratory

- Mentored a student on a research project related to neuromorphic chips and brain-inspired computing hardware

Technical Skills

Hardware Design and Simulation Tools: Synopsys Tools (HSPICE, Cscope, Design Compiler), Cadence Virtuoso Tools(Cadence Schematic XL, Cadence Layout EXL, Cadence Pegasus, Quantus, Innovus), Icarus Verilog, GTKWave

Scripting Languages: TCL

Hardware Description Language: Verilog

Programming Languages: C, C++, Python

Project Management Tools: Jira, Trello, Confluence

Engineering Tools: Visual Studio Code, Code Composer Studio, Keil MDK Compiler,IAR Compiler, CodeVision, STM32QubeMX, MATLAB, Altium Designer

Honors and Awards

2024 **Fanavaran DoorAndish Eshraq**, Recognized for Outstanding Performance and Dedication

2023 **Fanavaran DoorAndish Eshraq**, Recognized for Excellence in Detail Orientation and Accuracy

2020 **Islamic Azad University - Science and Research Branch, Tehran**, Ranked among the top 3 graduate students in the M.Sc. program

Publications

N. Mohajeri, B. Ebrahimi, and M. Dousti, “HPM: High-Precision Modeling of a Low-Power Inverter-Based Memristive Neural Network,” *Circuits, Systems, and Computers*, vol. 30, no. 15, 2021.